# JANGARA BLISS

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## EDUCATION

**Fort Lewis College — B.S. Computer Engineering** May 2027

Minors: Mathematics & Business Administration · GPA 3.65 · Upper-division GPA 3.8

## RESEARCH EXPERIENCE

**Research Assistant — Humanoid Robotics** May – Aug 2026 · paper in prep

Fort Lewis College · Advisor: Dr. Yiyan Li · Independent undergraduate research

- Integrated the Booster K1 into VLN-CE-Isaac / NaVILA and completed all **1,077** R2R val-unseen simulation episodes: **18.3% success**, **30.3% oracle success**, 10.9% SPL, and 7.59 m final navigation error.
- Decomposed outcomes into 751 never-reached, 129 reached-without-stopping, and 197 successful episodes, separating exploration from arrival recognition and motivating a controlled viewpoint study.
- Ran a preregistered **2,400-episode** viewpoint/transform sweep; all pre-specified contrasts were null, and identically configured runs disagreed on **11.0%** of episode outcomes — the evaluation-reliability result now in a first-author paper in preparation (AIRC 2027).
- Extended upstream K1 training/deployment surfaces with environment configuration, reward and observation-contract checks, wrappers, validators, command filtering, and safety-aware mode handling; canonical results include per-episode artifacts and a public evidence ledger.

**Independent Research — xembench** 2026

Language-grounded manipulation across a Franka arm and Unitree G1 in ManiSkill3

- Designed and executed a two-embodiment benchmark with a canonical 21-D action interface, 1,202 demonstrations, 18 BC policies, and a **73-cell / 6,550-episode** baseline matrix; reported near-zero transfer under weak native precision policies.
- Ran a failure-driven data round and intervention study; action chunking improved tool-pulling success from **1.3% to 14.7%** across three seeds, while grasp-critical tasks remained near zero. AI-assisted implementation is documented in the repository.

**Team Lead — NASA Colorado Robotics Challenge** Oct 2025 – Apr 2026

Fort Lewis College · Four-person team

- Led integration for an 18-DoF hexapod fielded at Great Sand Dunes; worked on Arduino/IMU/bump-sensor/power integration and a three-bit behavior interface, with team and inherited-code provenance stated publicly.
- Built a later Python Kuramoto coupled-oscillator gait simulator and 34-test CI suite; the simulator is presented separately from the physical challenge firmware.

**Research Assistant — AI & Robotics** May – Sep 2025

Fort Lewis College · Advisor: Dr. Kevin Wedeward · Solo project

- Restored legacy Sawyer and Baxter platforms, rebuilt a Dockerized ROS 1 + ROS 2 (Indigo/Humble) MoveIt and Gazebo workstation, and developed Blender/Python synthetic-data and YOLO inspection tooling; 2nd place, FLC Physics & Engineering Symposium 2025.

## SELECTED TECHNICAL WORK

**Bridge Crack-Inspection Arm — reBot B601-DM (Senior Design)** Sep 2026 – Present

Two-person team · Advisor: Dr. Yiyan Li

- Brought up 7 motors over CAN at **200 Hz** (MIT mode); measured the elbow gravity load (6.6 N·m vs the URDF's 2.7 N·m) and identified gravity/friction feed-forward on hardware, cutting loaded-joint hold error to ≤ **0.005 rad** in post-tuning holds across a 41-run logged campaign.
- Built the analytic Jacobian, DLS IK, and minimum-jerk planner behind a last-line safety guard; the first guarded trajectory tracked within 0.023 rad, and a wrist D405 runs zero-shot crack segmentation at ~17 Hz.

**KUKA KR 6 Kinematics & Verification** 2026

Individual robotics course project; later packaged as a public library

- Implemented FK, closed-form position IK, a 6×6 geometric Jacobian, singularity diagnostics, inverse velocity kinematics, and trajectories; verified with finite differences, 200 random round trips, a generated URDF, and **39 tests**.

## TECHNICAL SKILLS

**Languages:** Python, C++, Verilog, JavaScript / TypeScript, SQL, Bash

**Robot learning / AI:** PyTorch, behavior cloning, PPO, NaVILA, vision-language navigation, YOLO, experiment design

**Robotics:** Isaac Sim / Lab, ManiSkill3, MuJoCo, ROS / ROS 2, MoveIt, Gazebo, kinematics, motion planning

**Hardware / tools:** Arduino, FPGA / Vivado, PCB design, sensors, Git, Docker, Linux, AWS, PostgreSQL, Neo4j

## LEADERSHIP & HONORS

**Student Representative, FLC Strategic Implementation Committee** · **President & Co-Founder, Entrepreneurial Ventures Association** (8-person executive team; $1,500 in student micro-grants) · Physics & Engineering Symposium **2nd Place** (2025) · Dean's List (Spring 2026) · Goldman Sachs Emerging Leaders (2024)